

HENRY MARTINEZ

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EDUCATION

University of Southern California

Master of Science, Cyber Security Engineering
GPA: 4.0

In-progress

California State University, Northridge

Bachelor of Science, Computer Science
GPA: 3.65, Honors

Graduated May 2025

Relevant Coursework: Advanced Data Structures, Algorithms & Computer Programming, Calculus I & II, Computer Architecture & Organization, Data Structures & Program Design, Data Mining, Database Design, Discrete Structures, Intro Probability, Linear Algebra, Machine Learning for Data Science

SKILLS

Languages: Python (Advanced), Java (Intermediate), Javascript (Intermediate), C++ (Basic), SQL (Advanced), Bash (Intermediate)

Frameworks: Flask, scikit-learn, GeoPandas, PyTorch, AD/LDAP, Docker, Microsoft Services (Azure, PowerBI, Excel, Powerpoint, Word), Jira, Confluence

Domains: Identity and Access Management, Predictive Modeling, Sentiment Analysis, Data Visualization, Geospatial Data, Urban Policy, Environmental Policy

EXPERIENCE

NASA Jet Propulsion Laboratory

June 2023 - Present

Intern

- Developed Python web applications to automate single sign-on (SSO) request and integration processes, improving efficiency by ~97%.
- Standardized SSO integration processes, implementing automation, documentation, and status-tracking, eliminating manual work.
- Configured ~300 SSO integrations for on-premises and cloud applications, supporting over 5,500 employees.
- Migrated ~120 applications using a legacy SSO provider to a new NASA-compliant MFA provider, including NASA/JPL mission applications.
- Served as a primary contact for single sign-on inquiries, providing guidance and resolving technical issues for application owners.
- Developed real-time system monitoring/notification scripts and dashboards for ~10 critical IAM services used lab-wide.
- Improved our request intake process by creating a machine learning classification model that classifies labels for new requests, based on labels for ~20,000 previous tickets, with an accuracy score of roughly ~90%, improving efficiency by ~80%.
- Worked with NASA mission engineers on securing authentication infrastructure for missions such as NISAR, M2020, MSL, and Psyche.

NASA L'SPACE Proposal Writing and Evaluation Experience (NPWEE)

December 2022 - April 2023

Proposal Writer & Academic Researcher

- Collaborated with a nationwide team of 12 students on a NASA-guided research proposal, focused on generating solutions for space navigation and communication, with an emphasis on efficient and secure data transmission.
- Researched cybersecurity policies and procedures, brainstorming potential space-related vulnerabilities and proposing mitigation strategies.
- Developed skills in writing and reviewing existing policies, new technology reports, and proposals under the mentorship of subject matter experts.

PROJECTS

AI Altadena Fire Relief Models | Python, scikit-learn, Geospatial Analysis, Policy Analysis & Implementation

December 2025 - Present

- Assessing the competency of AI damage prediction models by comparing to official FEMA damage assessments.
- Uncovering whether AI models inadvertently undercount losses in lower-income and minority communities.

Are LA Public Libraries Equitably Placed? | Python, Geospatial Analysis, Policy Analysis & Implementation

July 2025 - December 2025

- Analyzed library placement equity across 256 LA County neighborhoods using geospatial representations of Census and library data.
- Calculated accessibility metrics and educational need indicators by neighborhood, utilizing neighborhood-level data.
- Uncovered a near-zero correlation between library placement and educational needs, suggesting a lack of equity-planning.
- Built a comprehensive web application with interactive maps and visualizations, outlining analysis steps and findings.
- Submitted policy recommendations to local government officials and library systems.

NASA Great Lunar Expedition for Everyone (GLEE) | C++, Arduino IDE, Teamwork

November 2022 - July 2023

- Collaborated with a team of six students from three state-wide universities on a NASA-funded project to program a miniature spacecraft using C++ and Arduino IDE, collecting local surface data for future lunar mission application and implementation.

ORGANIZATIONS & AFFILIATIONS

DataSC | Project Contributor

August 2025 - Present

- Actively participating in activities and projects related to machine learning, artificial intelligence, and natural language processing.

CERTIFICATES

Using Public Policy for Social Change | University of Michigan

December 2025